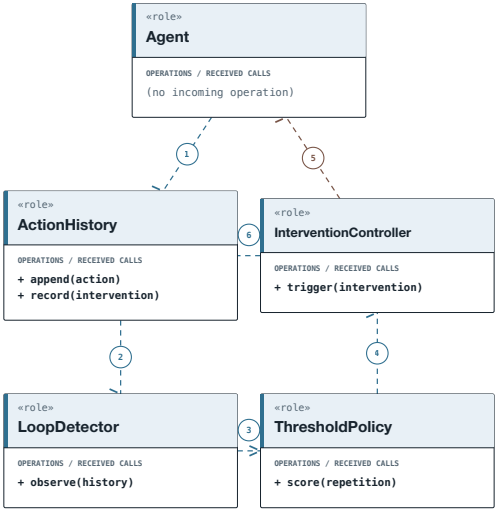
P18 Hybrid loop detection

Primitive / UML CLASS VIEW / ANALYTICAL MODEL

Detect stalled or repetitive behavior.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Successive agent observations or actions can repeat without progress.
TRIGGER	A loop signal crosses a configured threshold or composite condition.
STRUCTURE	Repetition signals, thresholds, and an intervention interface.
EFFECT	The harness diagnoses a loop and stops, redirects, or escalates.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Action or observation history is available in comparable form.
ENSURES	Intervention is attributed to explicit evidence and recorded in the trace.
INVARIANT	A detector cannot block all exploration solely because outputs are similar.
VIOLATION	Detector error, disabled mode, or false positive has a recoverable path.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 Agent → ActionHistory append(action)
2 ActionHistory → LoopDetector observe(history)
3 LoopDetector → ThresholdPolicy score(repetition)
4 ThresholdPolicy → InterventionController trigger(intervention)
5 InterventionController ↩ Agent stop_or_redirect()
6 InterventionController → ActionHistory record(intervention)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Action repetition, observation similarity, semantic signals, and composite thresholds.
TEST_OBLIGATION	Test exact repeats, productive variation, false positives, and intervention.
NEGATIVE_CONTROL	A recursion depth limit alone is not semantic loop detection.